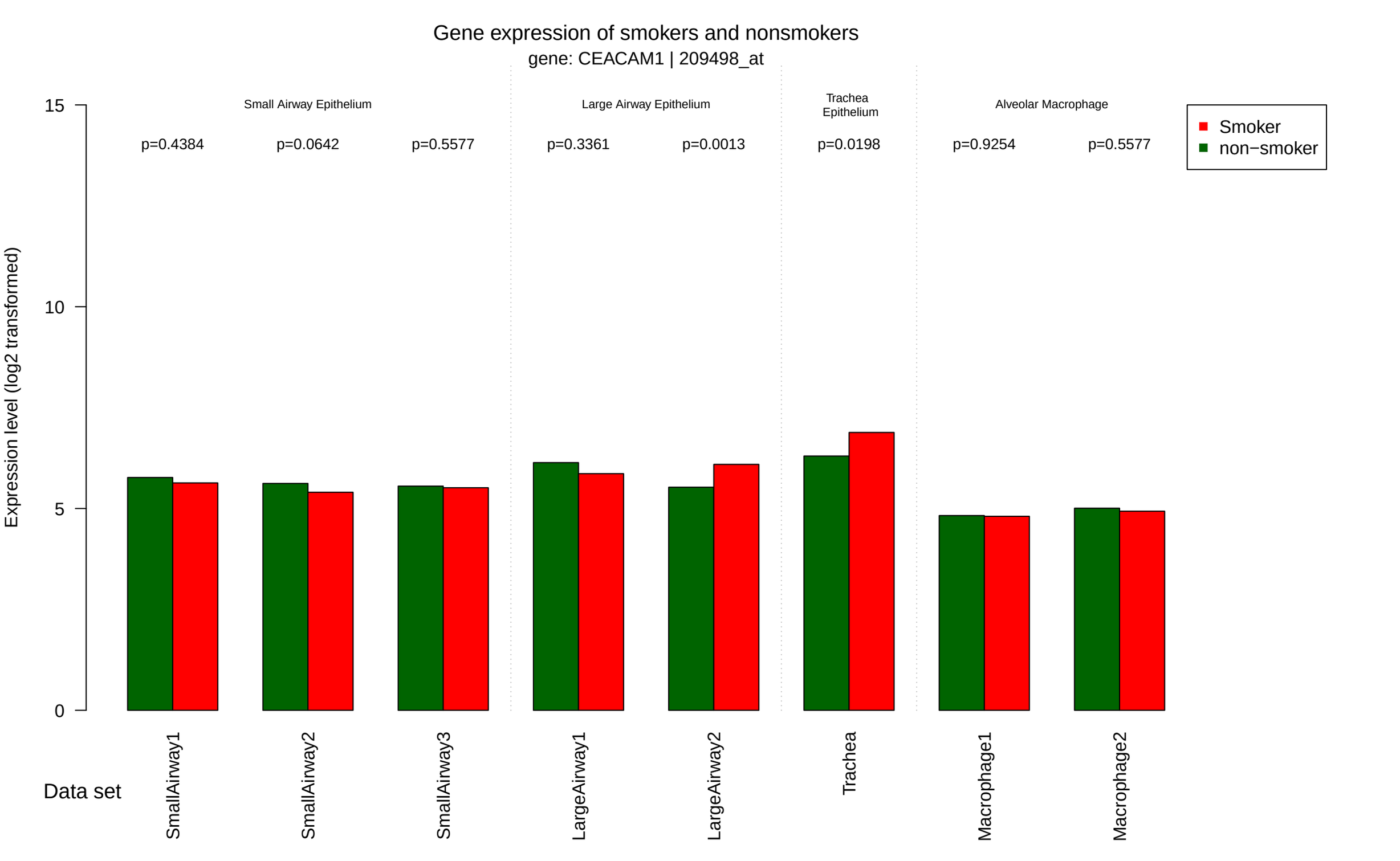
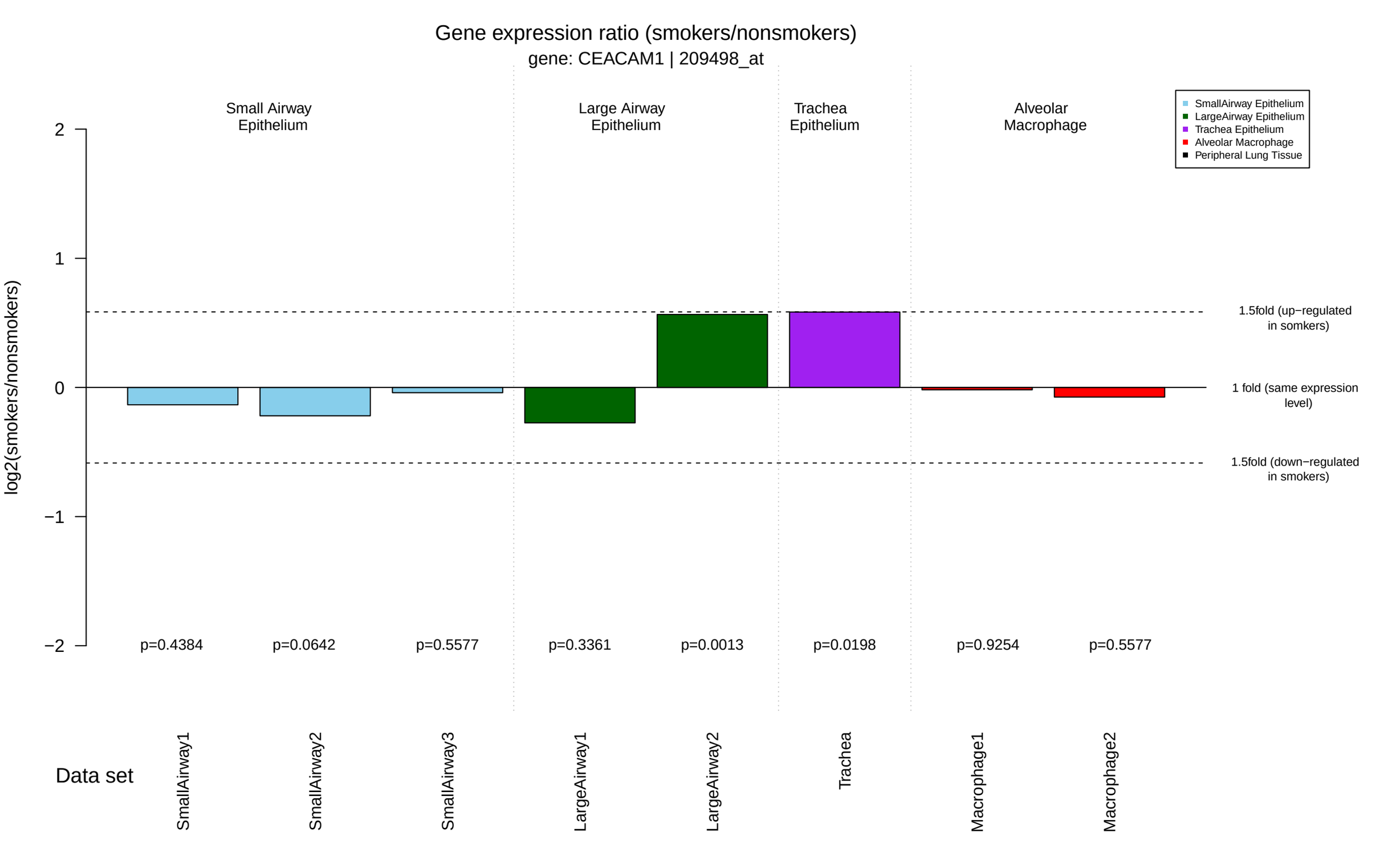
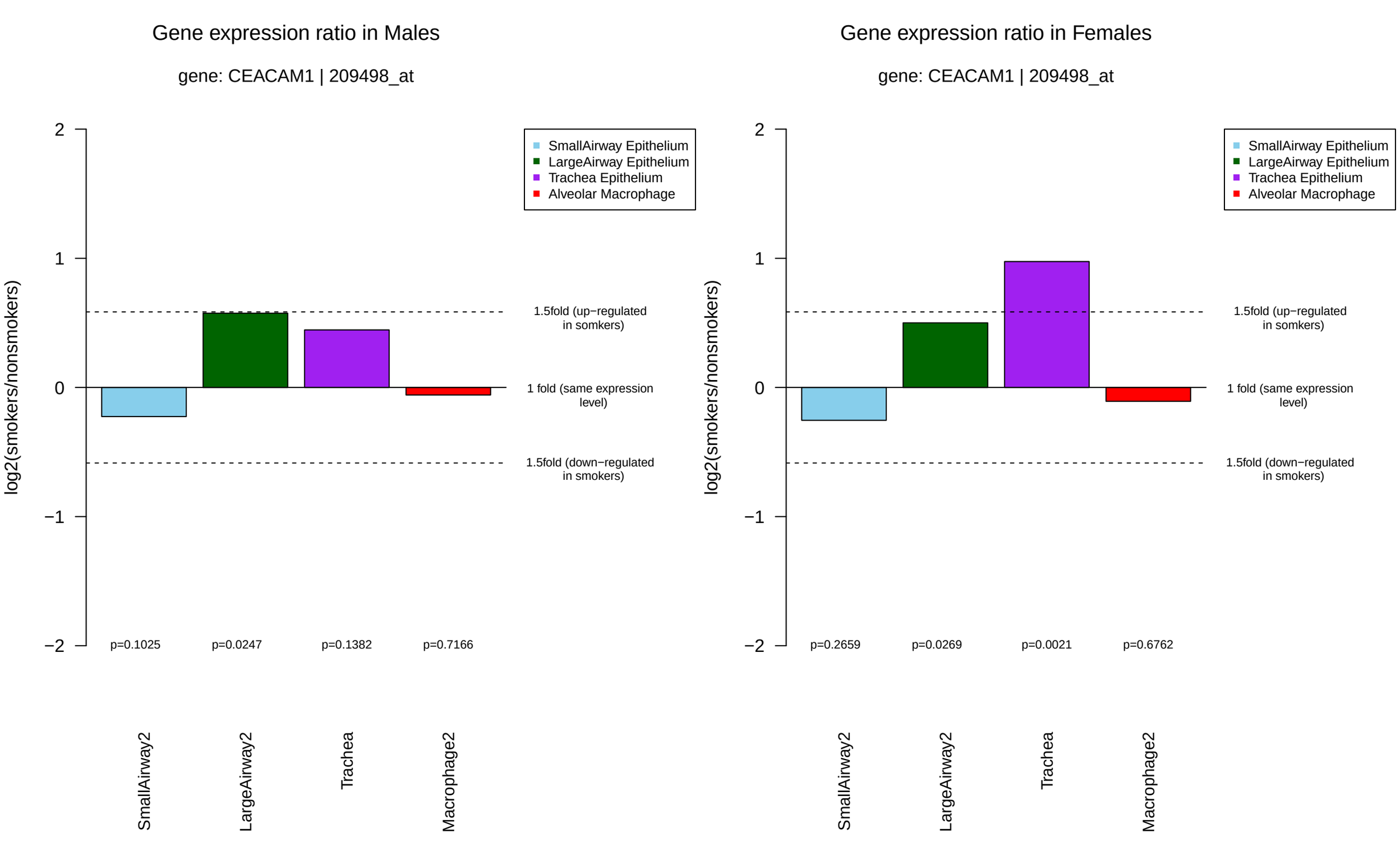
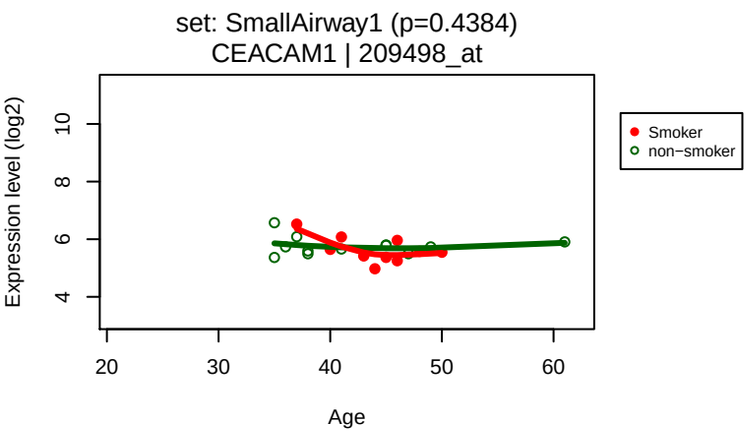
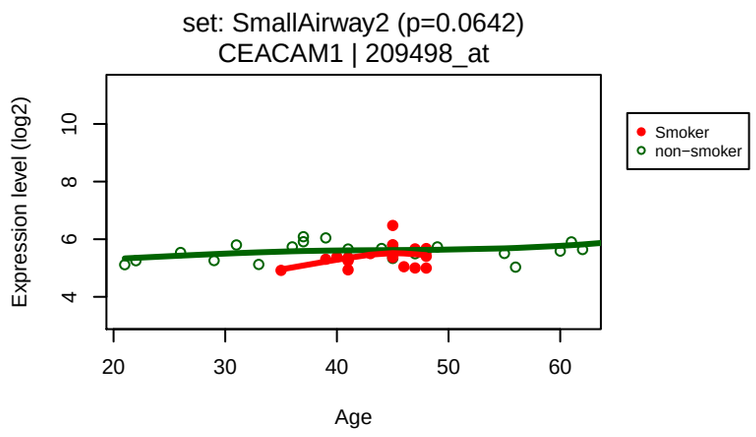
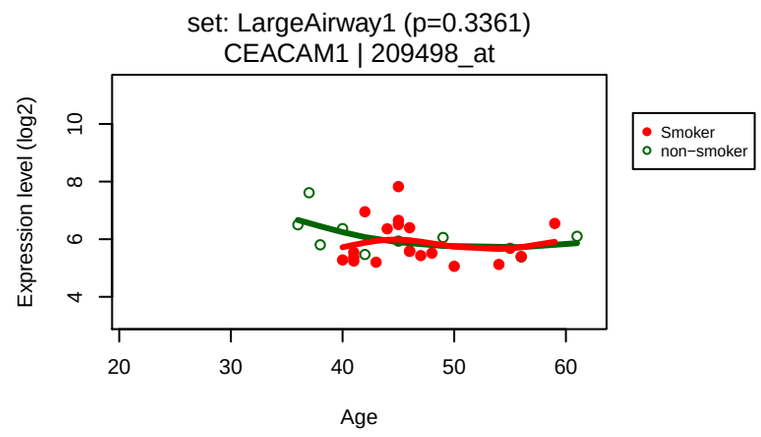

Gene expression of smokers and nonsmokers
gene: CEACAM1 | 209498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4384
p=0.0642
p=0.5577
p=0.3361
p=0.0013
p=0.0198
p=0.9254
p=0.5577
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

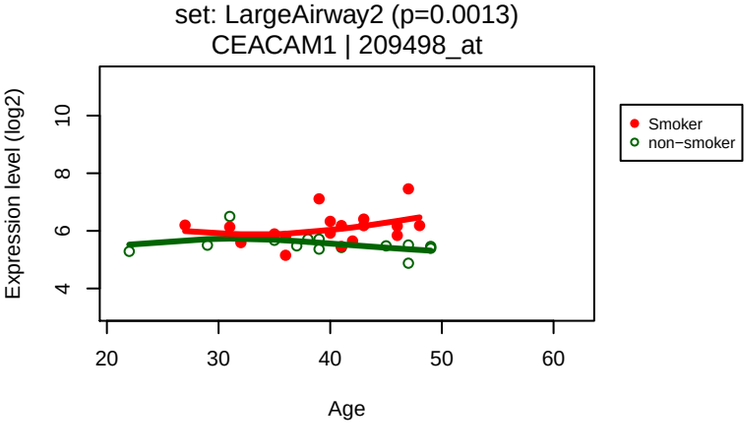

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM1 | 209498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4384
p=0.0642
p=0.5577
p=0.3361
p=0.0013
p=0.0198
p=0.9254
p=0.5577
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

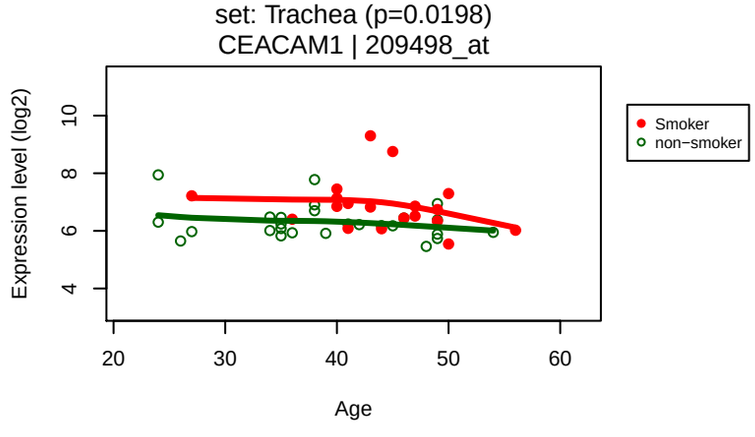

Gene expression ratio in Males
gene: CEACAM1 | 209498_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1025
p=0.0247
p=0.1382
p=0.7166
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM1 | 209498_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2659
p=0.0269
p=0.0021
p=0.6762
SmallAirway2
LargeAirway2
Trachea
Macrophage2

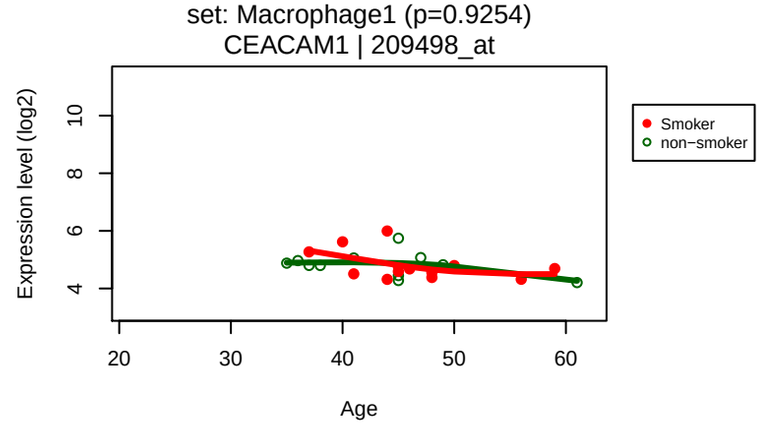

set: SmallAirway1 (p=0.4384)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker

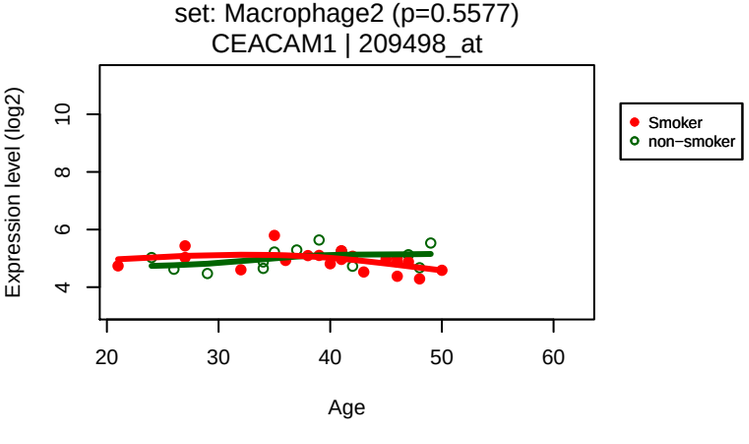

set: SmallAirway2 (p=0.0642)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker

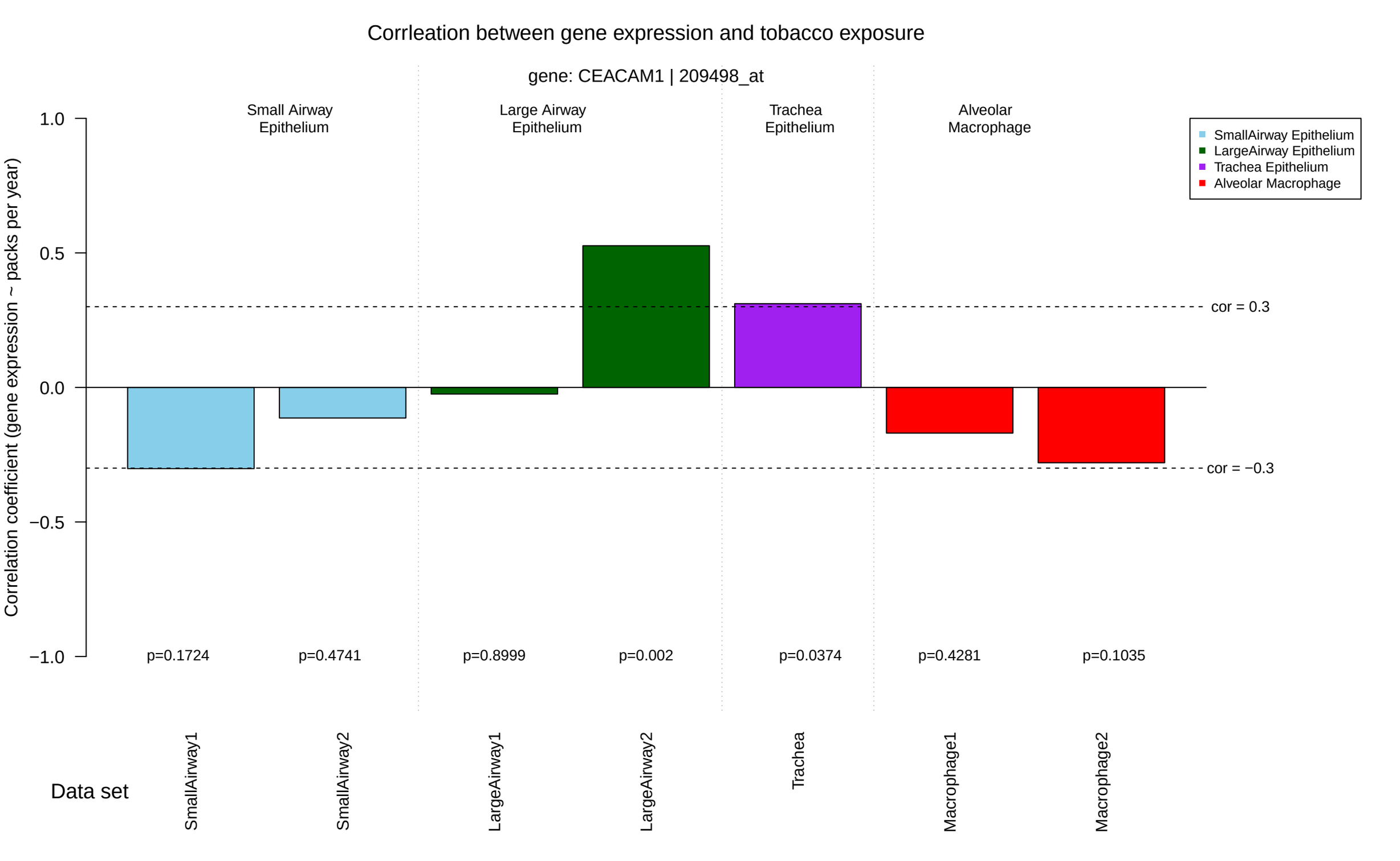

set: LargeAirway1 (p=0.3361)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0013)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0198)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9254)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5577)
CEACAM1 | 209498_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM1 | 209498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1724
p=0.4741
p=0.8999
p=0.002
p=0.0374
p=0.4281
p=0.1035
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2